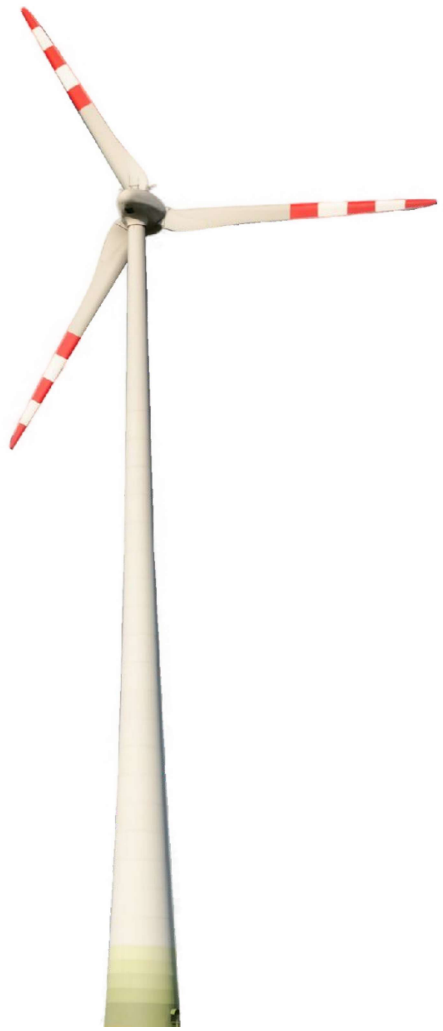
A Comprehensive Review of Data-Driven and Machine Learning-Based Fault Detection in Wind Turbines

CINE-CMD Workshop on AI for Materials and Energy



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Context and Motivations

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Wind energy is one of the major global renewable energy sources

Turbines operate in remote environments

Component failures[2]:
→ Unplanned downtime
→ High maintenance costs
→ Energy losses

Need for Early Fault Detection[3]:
→ Increase reliability
→ Reduce operational costs

1: Zhang, S., Robinson, E., & Basu, M. (2023). Wind Turbine Predictive Fault Diagnostics Based on a Novel Long Short-Term Memory Model. Algorithms, 16, 546. https://doi.org/10.3390/a16120546
García Márquez, F. P., & Pinar Pérez, J. M. (2020). Wind turbines: A general reliability analysis (pp. 1–18). Butterworth-Heinemann. https://doi.org/10.1016/B978-0-08-101094-5.00001-0
Santelo, T. N., Rodriges de Oliveira, C. M., Maciel, C. D., & Boffino de Almeida Monteiro, J. R. (2021). Wind Turbine Failures Review and Trends. Journal of Control, Automation and Electrical Systems, 1–17. https://doi.org/10.1007/S40313-021-00789-8

2:Puruncajas, B. (2024). Detection and diagnosis of faults and damage in wind turbines. https://doi.org/10.5821/dissertation-2117-407224

First Steps

Reading of relevant publications on the topic: *"Fault detection in wind turbines"*

PICO framework

Build the search string (below[1]) and define questions

Search in SCOPUS and IEEE Xplore:
276 results
(conferences and journals from 2015 to 2024-dez)

LLM's to suggest topic categories from title and abstract

Prompt for help:

Prompt: "Here are some articles with their respective titles and abstracts in JSON format. Please suggest topics or categories to group them under. The topic is "fault detection in wind turbines" and the search string that resulted in these papers was [generical search string].
Provide a brief description of each topic.
JSON:{
"index": "",
"Title": "",
"Abstract": ""
}"

[1]Generic search string covering the full context of the topic:

(("Wind Turbine" OR "Wind Energ*" OR "Wind Farm*") AND ("Machine Learning" OR "Artificial Intelligence" OR "Deep Learning" OR "Neural Network*" OR "ML" OR "AI") AND ("Fault Detection" OR "Fault Diagnosis" OR "Fault Prediction" OR "Failure Detection" OR "Condition Monitoring" OR "Predictive Maintenance" OR "Anomaly Detection" OR "Fault Prognosis" OR "Failure Prognosis") AND ("Performance" OR "Accuracy" OR "Efficiency" OR "Precision" OR "Recall" OR "Detection Rate" OR "Mean Absolute Error" OR "Mean Squared Error" OR "Reconstruction Error" OR "Cross-Entropy Loss" OR "AUC" OR "ROC" OR "F1-Score" OR "Root Mean Squared Error" OR "RMSE") AND ("SCADA" OR "Supervisory Control and Data Acquisition" OR "CMS" OR "Condition Monitoring System*" OR "Vibration Monitoring" OR "IoT-based Monitoring" OR "Remote Monitoring" OR "Sensor-based Monitoring" OR "Structural Health Monitoring" OR "SHM"))*

Legend:

- 🔴 In red: systems and structures related to wind energy
- 🟠 In orange: AI methods
- 🟡 In yellow: specific techniques
- 🟢 In green: performance evaluation metrics
- 🔵 In blue: data sources

First Insights

All LLM results

Prompt to condense the categories - Suggested final categories

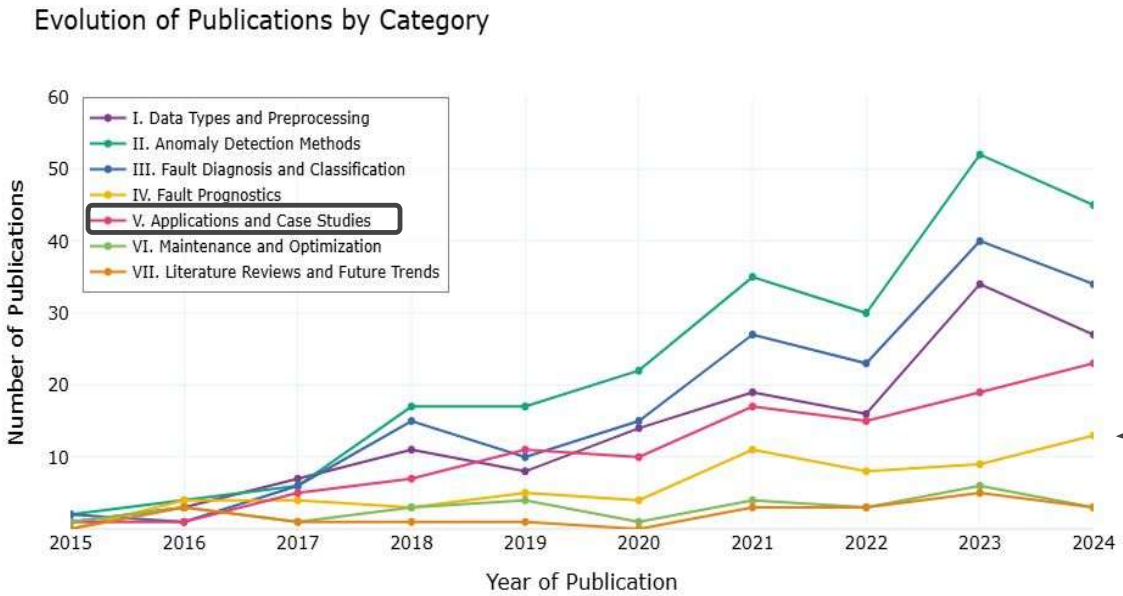
Seven main categories

LLM to Categorize each paper using Title and Abstract[1]

Result

Get New insights

Repeat for **"applications and case studies"**, and as a result we have the subcategories:



With this result, we can understand what is hot and go deeper into the topic of applications and case studies (Specific Components (Blades, Gearboxes, Generators, etc.), FOWT (offshore turbines), Newly Built Turbines).

Subcategories:

I - Blades
II - Gearboxes
III - Generators
IV - Main Shaft/Main Bearing/Spindle
V - Hydraulics/Pitch System
VI - Towers
VII - Foundations
VIII - FOWT (Floating Offshore Wind Turbines)
IX - Newly Built Turbines/Limited Data

They deal directly with the main components of wind turbines.

1: *Each paper can be in more than one category*

Structuring unstructured data

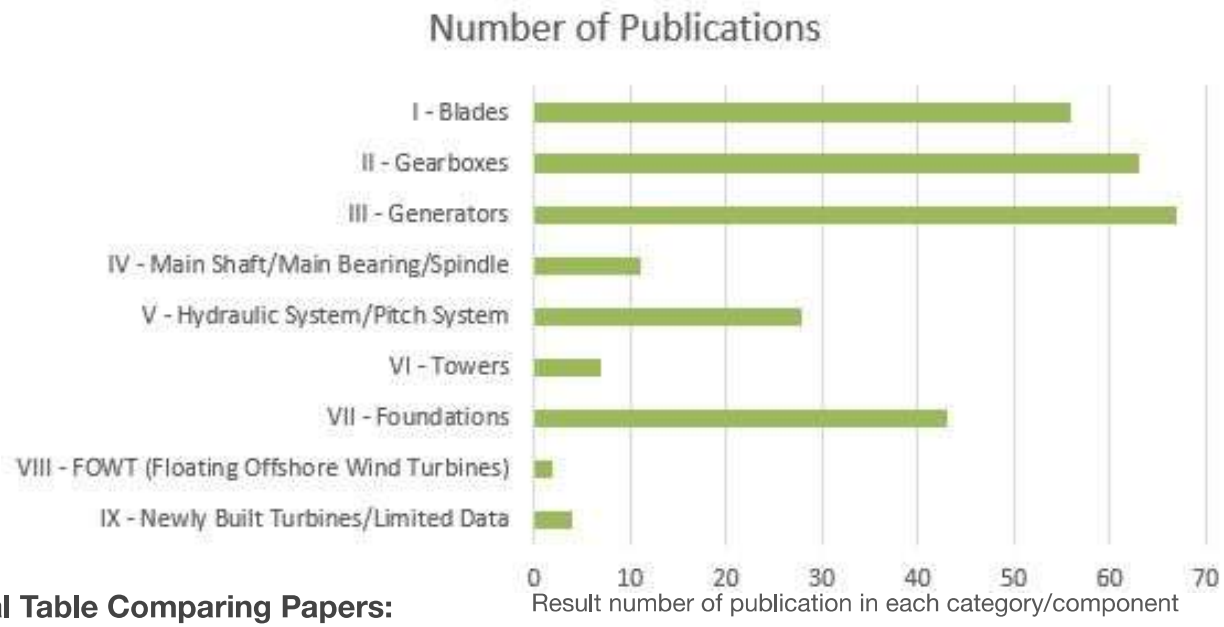
With Subcategories

Created a specific search query for each subcategories/component with the purpose of minimizing the biases introduced by LLM-assisted categorization

Search in SCOPUS and IEEE Xplore:

*[INITIAL SEARCH STRING] AND (SUBCATEGORY/**COMPONENT**))*

Final Papers



Help from LLM's to extract information and structure data from each .pdf

Result

Final Table Comparing Papers:

- Component
- Data type
- Real data ?
- ML Technique
- Task (Detection, prognosis,...)
- Metrics
- Main result
- Contributions
- Limitations
- Include (Yes/No)
- Justification

Analysis and discussion of subcategories

Summarize and identify gaps

Conclusions

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Identified Research Gaps and Proposed Improvements:

There is a lack of open datasets and standard benchmarks:
This makes it difficult to compare models fairly

Most works consider only single failures:
In practice, turbines may experience multiple interacting faults.
Also, causal variables are rarely included in the models.

Finally, very few studies perform cost-benefit or ROI analysis.

Key Challenge: The focus should not be only on model accuracy, but also on

- Cost reduction
- Increased availability
- Financial return

Emerging Directions:

- **Generative AI** for rare failure simulation
- **Reinforcement learning & multi-agent systems** for autonomous decisions
- **Multimodal models** combining SCADA, vibration, and thermal data

Main Goal:

Align AI research with real **industrial and financial impact.**

Thank You !